Insurance Claims Dataset - Variable Descriptions

Complete Data Dictionary

Dataset Overview

This document provides descriptions for all variables in the insurance claims dataset containing 114,321 claims with 133 columns (ID + target + 131 features).

Objective: Predict whether a claim is suitable for accelerated approval processing (target=1) versus standard processing (target=0).

Target Variable

target: Binary classification indicator for claim processing type. Value 0 indicates standard processing requiring normal review timeline, while value 1 indicates claims suitable for accelerated approval and fast-track processing.

Categorical Variables

claim_type: Type of insurance claim being filed, representing different categories such as auto, property, life, health, travel, liability, or disability claims.

claim_detail_code: Detailed claim classification code or product identifier with highly granular categorization combining multiple sub-attributes such as product type, sub-category, coverage level, and region code.

risk_category: Risk classification or underwriting category assigned to the claim, ranging from very low risk to critical risk, used for underwriting and pricing decisions.

policy_status: Current status of the insurance policy at time of claim, such as active/in-force, grace period, or lapsed/suspended.

geographic_region: Geographic region or territory where the claim originated, including classifications such as urban/metropolitan, suburban, or rural/remote areas.

document_completeness: Documentation completeness score or verification status, ranging from complete to incomplete with various review statuses.

payment_method: Requested payment or settlement method for the claim, such as direct deposit, check, card payment, third-party payment, or direct payment to provider.

coverage_type: Specific insurance coverage or benefit type being claimed, representing different insurance benefit products.

claim_channel: Channel through which the claim was submitted, such as online/digital portal, agent/broker, or phone/call center.

priority_level: Claim processing priority level where 1 indicates high priority/urgent, 2 indicates standard priority, and 3 indicates low priority/non-urgent.

fraud_indicator_flag: Automated fraud detection system flag indicating fraud risk levels from low to high, including flags for investigation.

customer_segment: Customer segment or policyholder classification, such as premium/VIP, standard, new, or at-risk customers.

claim_complexity: Assessed complexity level of the claim, ranging from very simple to highly complex claims requiring specialist review.

adjuster_recommendation: Claims adjuster's preliminary assessment or recommendation code, including approve, pending review, refer to specialist, or deny with various reasons.

automation_eligible: Flag indicating eligibility for automated processing, including fully automated, partially automated with oversight, or manual processing required.

provider_network_status: Status of service provider in network, indicating in-network, out-of-network, or preferred provider status.

claim_source: Source or origin of the claim, such as direct claim, subrogation, or third-party.

policy_type: Type of insurance policy under which claim is filed, such as auto insurance, basic life insurance, group insurance, universal life, or specialty insurance.

location_code: Geographic location code or processing center identifier, possibly representing regional or country codes.

Numeric Variables

Financial and Monetary Variables

claim_amount_ratio: Ratio of claim amount to policy coverage limit. Lower values indicate claims well within coverage limits; higher values approach policy maximums.

deductible_ratio: Ratio of deductible amount to claim amount, representing the proportion of claim covered by customer deductible.

premium_payment_score: Composite score based on premium payment history. Higher values indicate better payment track record.

claim_reserve_amount: Initial reserve amount set aside for the claim in currency thousands, representing expected claim payout estimate.

policy_value_score: Normalized policy value or sum assured score, indicating the relative value of the insurance policy.

total_claim_amount: Total claimed amount in thousands of currency units, representing the monetary value being claimed by the policyholder.

coverage_limit: Maximum coverage limit for this policy type in thousands, representing the upper bound of potential claim payment.

previous_claims_value: Total value of previous claims by this policyholder in thousands, showing historical claim amount indicator.

estimated_settlement_amount: Estimated final settlement amount in thousands, representing expected payout after adjustments and negotiations.

policy_premium_annual: Annual premium amount for the policy, representing the policyholder's annual payment commitment.

claim_to_premium_ratio: Ratio of current claim amount to annual premium, measuring claim severity relative to premium paid.

incurred_loss_estimate: Estimated total incurred loss including reserves, representing the total expected financial impact of the claim.

salvage_value: Estimated salvage or recovery value if applicable, representing potential reduction in net claim cost.

deductible_amount: Actual deductible amount in currency units, representing out-of-pocket expense for policyholder.

reinsurance_amount: Amount to be recovered from reinsurance, representing the portion of claim covered by reinsurer.

Temporal Variables

days_since_policy_start: Years since policy inception, serving as a policy tenure or age indicator.

months_since_last_claim: Months elapsed since previous claim if any, serving as a claim frequency indicator showing recency of previous claim.

claim_processing_days: Expected or typical processing days for similar claims, providing a processing timeline estimate.

policy_renewal_months: Months until next policy renewal, indicating position within policy cycle.

days_claim_to_report: Number of days between incident and claim reporting, measuring timeliness of claim submission and serving as a potential fraud indicator.

customer_tenure_years: Years the customer has been with the institution, measuring customer loyalty and relationship duration.

incident_to_claim_hours: Hours between incident occurrence and claim filing, measuring immediacy of claim reporting.

claim_received_to_review_hours: Hours from claim receipt to initial review, serving as an internal processing efficiency metric.

days_in_current_status: Days the claim has been in its current processing status, serving as a processing timeline tracker.

policy_age_days: Age of the policy in days at time of claim, serving as a policy maturity indicator.

Risk and Scoring Variables

risk_score: Automated underwriting risk score providing overall risk assessment where higher values may indicate higher risk.

fraud_probability_score: ML-based fraud probability score indicating likelihood of fraudulent claim.

credit_score_normalized: Customer credit score normalized to 0-10 scale, serving as a financial reliability indicator.

claim_complexity_score: Automated complexity assessment score predicting the difficulty of claim evaluation.

straight_through_processing_score: Probability of successful automated processing, measuring automation suitability.

underwriting_confidence: Confidence level in automated decision, measuring certainty of risk assessment.

customer_loyalty_score: Composite customer loyalty or retention score, measuring customer relationship strength.

medical_necessity_score: Medical necessity rating for health claims, providing clinical appropriateness assessment.

loss_ratio_history: Customer's historical loss ratio comparing claims to premiums, serving as a profitability indicator for this customer.

claim_approval_probability: Model-predicted probability of claim approval, indicating likelihood of successful claim.

severity_index: Claim severity index providing injury or damage severity classification.

predictive_risk_score: Forward-looking risk score providing future claim probability assessment.

settlement_confidence_score: Confidence in settlement amount accuracy, measuring estimation certainty.

anti_fraud_score: Anti-fraud system composite score providing fraud detection output.

claim_validity_score: Overall claim validity assessment serving as a legitimacy indicator.

customer_satisfaction_score: Historical customer satisfaction rating serving as a service quality metric.

prior_denial_indicator: Number of previous denied claims, showing historical denial pattern.

policy_compliance_score: Policy terms compliance rating measuring adherence to policy conditions.

Count and Frequency Variables

previous_claims_count: Number of previous claims by this policyholder, tracking claim frequency history.

policy_modifications_count: Number of times policy has been modified or amended, serving as a policy stability indicator.

documents_submitted_count: Number of supporting documents submitted with claim, measuring documentation thoroughness.

third_party_involved_count: Number of third parties involved in the claim, serving as a claim complexity factor.

witnesses_count: Number of witnesses listed if applicable, measuring claim substantiation level.

medical_providers_count: Number of medical providers involved in health claims, indicating treatment complexity.

line_items_count: Number of individual items or services claimed, measuring claim detail granularity.

policy_riders_count: Number of additional riders or endorsements on policy, counting coverage enhancements.

communication_touchpoints: Number of communications with customer regarding claim, measuring interaction frequency.

external_verifications_count: Number of external verifications required or completed, measuring validation effort level.

adjustments_count: Number of claim amount adjustments made, tracking claim refinement iterations.

extensions_requested: Number of deadline extensions requested, serving as a processing delay indicator.

appeals_filed_count: Number of appeals or disputes filed by the customer, measuring claim contestation level.

Ratio and Percentage Variables

coverage_utilization_rate: Percentage of total coverage utilized, measuring extent of coverage usage.

coinsurance_percentage: Co-insurance percentage applicable to claim, defining the cost-sharing ratio.

network_discount_rate: Discount rate for in-network providers in percentage, measuring cost savings from network usage.

depreciation_rate: Applicable depreciation rate for asset claims in percentage, serving as a value reduction factor.

recovery_probability: Probability of subrogation recovery, indicating expected recovery likelihood.

settlement_acceptance_probability: Likelihood customer will accept settlement offer, serving as a settlement negotiation indicator.

liability_percentage: Percentage of liability attributed to insured party, defining fault or responsibility allocation.

policy_lapse_probability: Risk of policy cancellation post-claim, measuring retention risk.

claim_to_limit_ratio: Ratio of claim amount to policy limit, measuring coverage adequacy.

claim_closure_rate: Historical rate of claims successfully closed, serving as a resolution efficiency metric.

payment_timeliness_ratio: Ratio of on-time premium payments, measuring payment behavior consistency.

benefit_utilization_rate: Historical rate of benefit utilization, measuring usage pattern indicator.

Geographic and Demographic Variables

regional_claim_frequency_index: Regional normalized claim frequency serving as a geographic risk indicator.

zip_code_risk_score: Risk score based on geographic location, providing area-based risk factor.

urban_density_index: Population density metric for claim location, indicating urban versus rural area.

regional_cost_index: Cost-of-living adjustment for region, measuring geographic cost variation.

distance_to_provider: Distance from customer to service provider in miles or kilometers, serving as an accessibility metric.

regional_fraud_index: Historical fraud rate for the region, measuring geographic fraud risk.

customer_age: Age of the policyholder serving as a demographic risk factor.

household_size: Number of people in policyholder's household, indicating family structure.

income_bracket_index: Normalized income level indexed as a socioeconomic indicator.

Technical and System Variables

data_completeness_score: Percentage of required data fields completed, serving as a data quality metric.

system_confidence_score: Automated system's confidence in recommendation, measuring decision certainty.

duplicate_claim_similarity: Similarity score to potential duplicate claims, serving as a duplicate detection metric.

image_quality_score: Quality assessment of submitted photos or documents, measuring evidence quality.

ocr_confidence_score: OCR text recognition confidence for document processing, measuring document processing accuracy.

data_source_reliability: Reliability score of claim data source, measuring information trustworthiness.

model_version_indicator: Numeric indicator of ML model version used for scoring, serving as a system version tracker.

processing_queue_priority: Numeric priority in processing queue, indicating processing order.

sla_compliance_score: Service level agreement compliance metric measuring processing timeliness.

system_latency_score: Processing time score normalized, measuring system performance.

api_response_quality: Quality score of external API responses, measuring external data integration quality.

workflow_automation_score: Degree of automation in claim workflow, measuring automation coverage.

Medical and Health-Related Variables

diagnosis_complexity_score: Complexity of medical diagnosis for health claims, measuring clinical complexity.

treatment_cost_index: Indexed treatment cost relative to regional averages, measuring cost appropriateness.

recovery_time_estimate: Estimated recovery or treatment duration in days, measuring treatment timeline.

medical_necessity_index: Medical necessity composite index, measuring clinical appropriateness.

provider_quality_score: Quality rating of medical provider, measuring provider performance.

comorbidity_score: Number or severity of co-existing medical conditions, measuring patient complexity.

Policy and Product Variables

policy_tenure_score: Normalized policy tenure score, measuring policy longevity.

premium_to_coverage_ratio: Ratio of premium to coverage amount, measuring policy efficiency.

policy_persistency_score: Policy persistency or renewal rate, measuring policy stability.

coverage_breadth_index: Number or extent of different coverages, measuring policy comprehensiveness.

policy_flexibility_score: Flexibility and customization level of policy, measuring product adaptability.

cross_sell_index: Number of additional products held by customer, measuring customer relationship depth.

premium_payment_frequency: Payment frequency indicator for premiums, measuring payment pattern.

policy_conversion_indicator: Flag for converted policies from other types, measuring policy transformation.

benefit_enhancement_score: Score for additional benefits or riders, measuring enhanced coverage level.

policy_age_at_claim_ratio: Ratio of policy age to expected policy term, measuring policy lifecycle position.

coverage_gap_indicator: Indicator of gaps in coverage, measuring coverage adequacy.

policy_upgrade_count: Number of policy upgrades or enhancements, measuring policy evolution.

premium_discount_rate: Applicable discount rate on premiums, measuring pricing concessions.

Behavioral and Interaction Variables

claim_filing_pattern_score: Score based on claim filing behavior patterns, measuring behavioral consistency.

customer_engagement_score: Level of customer engagement with company, measuring relationship activity.

digital_adoption_index: Usage of digital channels and tools, measuring digital preference.

complaint_history_score: Historical complaint frequency and severity, measuring service satisfaction.

Derived and Composite Variables

accelerated_processing_indicator: Composite indicator for accelerated processing eligibility, combining multiple factors for fast-track determination.

final_decision_confidence: Overall confidence in the processing decision, combining all assessment factors into a final confidence score.

Summary

This data dictionary provides complete descriptions for all 131 feature variables in the insurance claims dataset. Variables are organized into logical categories to facilitate understanding and analysis:

- **1 Target Variable:** Binary outcome for accelerated processing
- **19 Categorical Variables:** Claim, policy, and customer classifications
- **112 Numeric Variables:** Spanning financial, temporal, risk, frequency, demographic, technical, medical, policy, and behavioral domains

All variable names represent descriptive interpretations of the original coded variables (v1-v131) to enhance model interpretability and business understanding.